Module 6: Metabolism - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Apply thermodynamic ideas to living cells.
2. Explain ATP as an energy-coupling molecule.
3. Describe how enzymes change reaction rates.
4. Predict how environmental conditions affect enzyme activity.
5. Connect photosynthesis and cellular respiration in matter and energy flow.

Topics

- Energy and thermodynamics
- ATP as cellular coupling
- Enzymes and activation energy
- Regulation by conditions and inhibitors
- Photosynthesis and respiration as linked pathways

Key Terms to Know

- **Metabolism** - All chemical reactions in a cell or organism.
- **ATP** - Cellular energy-coupling molecule.
- **Activation energy** - Energy required to start a reaction.
- **Enzyme** - Protein or RNA catalyst that speeds a reaction.
- **Substrate** - Reactant that binds an enzyme active site.
- **Denaturation** - Loss of shape that disrupts function.
- **Photosynthesis** - Process that uses light energy to build sugars.
- **Cellular respiration** - Process that harvests usable energy from fuel molecules.

Core Contents

1. Cells transform energy while obeying thermodynamic constraints.
2. ATP couples energy-releasing reactions to energy-requiring cellular work.
3. Enzymes lower activation energy without being consumed.
4. Temperature, pH, substrate concentration, and inhibitors alter enzyme function.
5. Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-06_metabolism.md`

Generated Visuals

- Module 06: Metabolism Evidence Map
- Module 06: Metabolism Reasoning Sequence
- Module 06: Metabolism Retrieval and Lab Check